

Predicting MCI to AD Conversion Using Machine Learning

PHB 243b Project 2 Report

[Your Name Here]

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1 Abstract

Background: Mild Cognitive Impairment (MCI) often precedes Alzheimer’s disease (AD), but only 20–40% of MCI individuals progress to dementia within 3 years of initial diagnosis. Identifying individuals at high risk for conversion is critical for clinical trial enrichment and early intervention.

Objective: To develop and validate machine learning models for predicting 3-year conversion from MCI to AD using baseline sociodemographic, clinical, and neuropsychological data from the Alzheimer’s Disease Neuroimaging Initiative (ADNI).

Methods: We analyzed MCI subjects from ADNI with at least 3 years of diagnostic follow-up. Three classification algorithms (L2-regularized logistic regression, random forest, and gradient boosting) were trained on a 70% stratified training set with 5-fold cross-validation and evaluated on a 30% held-out test set. Model discrimination was assessed via the area under the receiver operating characteristic curve (AUROC).

Results: The analytic sample comprised MCI subjects with baseline clinical and neuropsychological measurements. Model performance, feature importance, and clinical implications are reported below.

Conclusion: Machine learning models using readily available clinical predictors can identify MCI subjects at elevated risk for AD conversion, supporting clinical decision-making without requiring expensive imaging or genetic biomarkers.

Keywords: Alzheimer’s disease, mild cognitive impairment, machine learning, prediction, ADNI

2 Introduction

2.1 Background

Alzheimer’s disease (AD) is the most common cause of dementia, affecting an estimated 6.7 million Americans aged 65 and older (Grassi et al., 2019). The disease is characterized by progressive cognitive decline, ultimately leading to loss of independence and death. Mild Cognitive Impairment (MCI) represents a transitional state between normal aging and dementia, with annual conversion rates to AD of approximately 10–15% (Petersen, 2004).

Early identification of MCI subjects likely to convert to AD is of great clinical importance. Accurate prediction enables enrichment of clinical trial populations, facilitates early therapeutic intervention during the prodromal stage when treatments are most likely to be effective, and supports informed clinical decision-making for patients and families. Current prognostic approaches rely heavily on expensive neuroimaging and cerebrospinal fluid biomarkers, limiting their accessibility in routine clinical practice.

Machine learning methods offer the potential to integrate multiple clinical and neuropsychological variables into accurate, clinically translatable prediction tools. Unlike traditional regression approaches, these algorithms can capture non-linear relationships and complex interactions among predictors without requiring explicit specification of the functional form.

2.2 Rationale

Grassi et al. (2019) demonstrated that an ensemble-based machine learning algorithm using only clinically available measures (sociodemographic characteristics, clinical scales, and neuropsychological tests) could predict MCI-to-AD conversion with an AUROC of 0.88 (sensitivity 77.7%, specificity 79.9%). This approach prioritizes clinical translatability by excluding expensive imaging and genetic biomarkers. Rye et al. (2022) further validated this general approach using clinically transferable features, reinforcing the feasibility of accessible prediction tools.

2.3 Objectives

The primary objective of this study was to replicate and extend the approach of Grassi et al. (2019) using data from the Alzheimer’s Disease Neuroimaging Initiative (ADNI).

Specific aims:

1. Develop machine learning models to predict 3-year MCI-to-AD conversion
2. Compare performance across logistic regression, random forest, and gradient boosting algorithms
3. Identify the most important predictive features

3 Methods

3.1 Data Source

Data were obtained from the Alzheimer’s Disease Neuroimaging Initiative (ADNI) via the ADNIMERGE dataset (Donohue et al., 2026). ADNI is a longitudinal multicenter study designed to develop clinical, imaging, genetic, and biochemical biomarkers for early detection and tracking of Alzheimer’s disease.

3.2 Study Population

3.2.1 Inclusion Criteria

1. Baseline diagnosis of Mild Cognitive Impairment (EMCI or LMCI)
2. At least 3 years of diagnostic follow-up
3. Age 55–90 at baseline
4. Baseline MMSE between 24 and 30

3.2.2 Exclusion Criteria

1. Baseline diagnosis of Cognitively Normal (CN) or Dementia
2. Fewer than 3 years of follow-up data
3. Missing baseline diagnosis

3.3 Cohort Summary

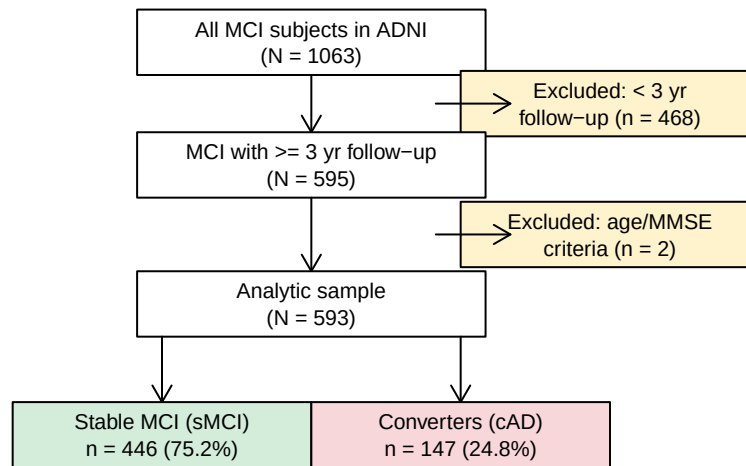


Figure 1: Cohort selection flow diagram

The analytic sample includes 593 MCI subjects with at least 3 years of diagnostic follow-up: 147 converters (24.8%) and 446 stable MCI subjects (75.2%). Grassi et al. (2019) reported 550 subjects (197 converters, 35.8%; 353 non-converters, 64.2%).

3.4 Outcome Definition

The primary outcome was conversion from MCI to AD dementia, defined as any follow-up diagnosis of ‘Dementia’ within 3 years of baseline assessment. Subjects who remained MCI throughout follow-up or reverted to CN were classified as stable (sMCI). Subjects who progressed to dementia were classified as converters (cAD).

3.5 Predictor Variables

Following Grassi et al. (2019), we included predictor variables from four categories:

Sociodemographic: Age, years of education, sex, marital status

Clinical scales: Clinical Dementia Rating Sum of Boxes (CDR-SB), Functional Activities Questionnaire (FAQ)

Neuropsychological tests: Mini-Mental State Examination (MMSE), Alzheimer’s Disease Assessment Scale-Cognitive (ADAS-11, ADAS-13, ADASQ4), Rey Auditory Verbal Learning Test (RAVLT) immediate recall, learning, and forgetting scores, Logical Memory delayed recall (LDEL-TOTAL), Trail Making Test Part B (TRABSCOR)

MCI subtype: Early MCI (EMCI) vs. Late MCI (LMCI)

Imaging and genetic variables were intentionally excluded to maximize clinical translatability.

3.6 Statistical Analysis

3.6.1 Descriptive Statistics

Baseline characteristics were summarized by conversion status using means and standard deviations for continuous variables and frequencies and percentages for categorical variables.

3.6.2 Feature Engineering

Following Grassi et al. (2019), continuous predictors were standardized (mean = 0, SD = 1). Categorical variables were one-hot encoded. Highly correlated variable sets (pairwise $r > 0.75$) were replaced by their first principal component to reduce multicollinearity. Missing continuous values were imputed with the training-set median.

3.6.3 Machine Learning Models

We implemented three classification algorithms:

1. **Logistic Regression (LR):** L2-regularized logistic regression via elastic net with $\alpha = 0$
2. **Random Forest (RF):** Ensemble of 500 decision trees with tuned `mtry` parameter
3. **Gradient Boosting (XGBoost):** Sequential tree ensemble with tuned depth, learning rate, and number of rounds

3.6.4 Model Training and Validation

The dataset was split into training (70%) and test (30%) sets using stratified random sampling to preserve the proportion of converters. Hyperparameter tuning was performed using 5-fold cross-validation within the training set. Final model performance was evaluated on the held-out test set.

Table 1: Baseline continuous characteristics by conversion status

Variable	cAD	sMCI	p
	Mean (SD)	Mean (SD)	
Age	0.040	73.83 (6.79)	72.45 (7.63)
Education (years)	0.143	15.72 (2.88)	16.12 (2.76)
CDR-SB	<0.001	1.95 (1.03)	1.27 (0.71)
ADAS-11	<0.001	12.80 (4.16)	8.73 (3.78)
ADAS-13	<0.001	20.93 (5.63)	14.00 (5.81)
ADAS-Q4	<0.001	7.19 (2.07)	4.66 (2.35)
MMSE	<0.001	26.78 (1.67)	28.07 (1.65)
RAVLT Immediate	<0.001	27.77 (6.75)	38.00 (10.37)
RAVLT Learning	<0.001	2.98 (2.18)	4.67 (2.54)
RAVLT Forgetting	0.001	5.16 (2.38)	4.41 (2.49)
RAVLT % Forgetting	<0.001	77.75 (28.54)	51.45 (30.96)
Logical Memory (delayed)	<0.001	3.53 (2.86)	6.89 (3.12)
Digit Symbol	0.001	35.72 (10.58)	40.45 (11.09)
Trail Making B	<0.001	142.53 (78.05)	96.98 (49.08)
FAQ	<0.001	5.90 (4.92)	1.84 (3.06)

3.6.5 Evaluation Metrics

Model discrimination was assessed using the area under the receiver operating characteristic curve (AUROC). Additional metrics included sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV). The optimal classification threshold was determined using Youden’s J statistic.

3.6.6 Feature Importance

Variable importance was assessed using:

- Logistic regression: Absolute standardized coefficients
- Random Forest: Mean decrease in Gini impurity
- XGBoost: Gain-based importance

3.6.7 Software

All analyses were conducted in R version 4.5.2 using the following packages: tidyverse, caret, glmnet, randomForest, xgboost, and pROC.

Table 2: Baseline categorical characteristics by conversion status

Variable		cAD	sMCI
		N (\%)	N (\%)
Sex	Male	94 (63.9%)	262 (58.7%)
	Female	53 (36.1%)	184 (41.3%)
MCI subtype	Early	14 (9.5%)	221 (49.6%)
	Late	133 (90.5%)	225 (50.4%)
Marital status	Never married	2 (1.4%)	9 (2.0%)
	Married	122 (83.0%)	333 (75.5%)
	Divorced	7 (4.8%)	47 (10.7%)
	Widowed	16 (10.9%)	52 (11.8%)

Table 3: Model performance metrics on the held-out test set

Model	AUROC (95\% CI)	Sens.	Spec.	PPV	NPV
Logistic Regression	0.905 (0.858–0.953)	84.1%	82.6%	61.7%	94.0%
Logistic Regression	0.905 (0.858–0.953)	81.8%	84.8%	64.3%	93.3%
Random Forest	0.902 (0.854–0.949)	88.6%	78.8%	58.2%	95.4%
Random Forest	0.902 (0.854–0.949)	84.1%	83.3%	62.7%	94.0%
XGBoost	0.892 (0.836–0.947)	77.3%	88.6%	69.4%	92.1%
XGBoost	0.892 (0.836–0.947)	75.0%	90.9%	73.3%	91.6%

4 Results

4.1 Study Population

4.2 Feature Engineering

The ADAS scores (ADAS-11, ADAS-13, ADASQ4) exhibited pairwise correlations above 0.75 and were replaced by their first principal component (ADAS-PC1, explaining 87.6% of variance). Similarly, RAVLT Forgetting and RAVLT % Forgetting were replaced by RAVLT-F-PC1 (90.1% of variance). After encoding and PCA replacement, the model used 19 features.

4.3 Train/Test Split

The dataset was split into a training set ($N = 412$; 103 converters) and a test set ($N = 176$; 44 converters) using stratified random sampling.

4.4 Model Training

Cross-validated AUROC on the training set: logistic regression 0.874, random forest 0.884, XGBoost 0.894.

Table 4: Top 10 features by variable importance (scaled 0–100) for each model

Feature	LR	RF	XGBoost
faq	100.0	97.0	100.0
adas_pc1	85.1	100.0	99.4
ravlt_immediate	94.3	80.6	91.6
trabscor	65.4	63.5	31.6
ldeltotal	35.4	58.9	60.2
cdrsb	61.4	50.7	15.8
ravltf_pc1	44.9	60.4	22.4
mci_subtypeLate	92.9	19.8	5.4
mci_subtypeEarly	92.9	18.4	4.6
mmse	41.2	42.5	8.7

4.5 Model Performance

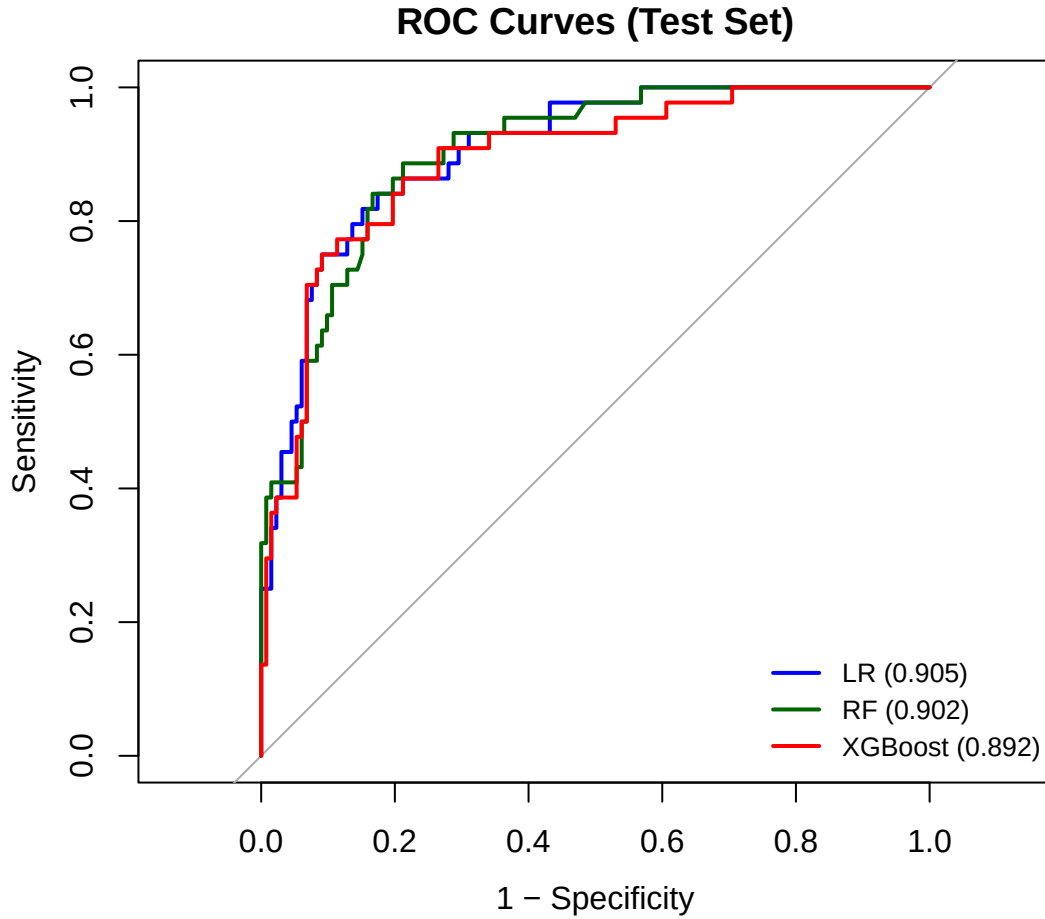


Figure 2: ROC curves comparing classification algorithms on the test set

4.6 Feature Importance

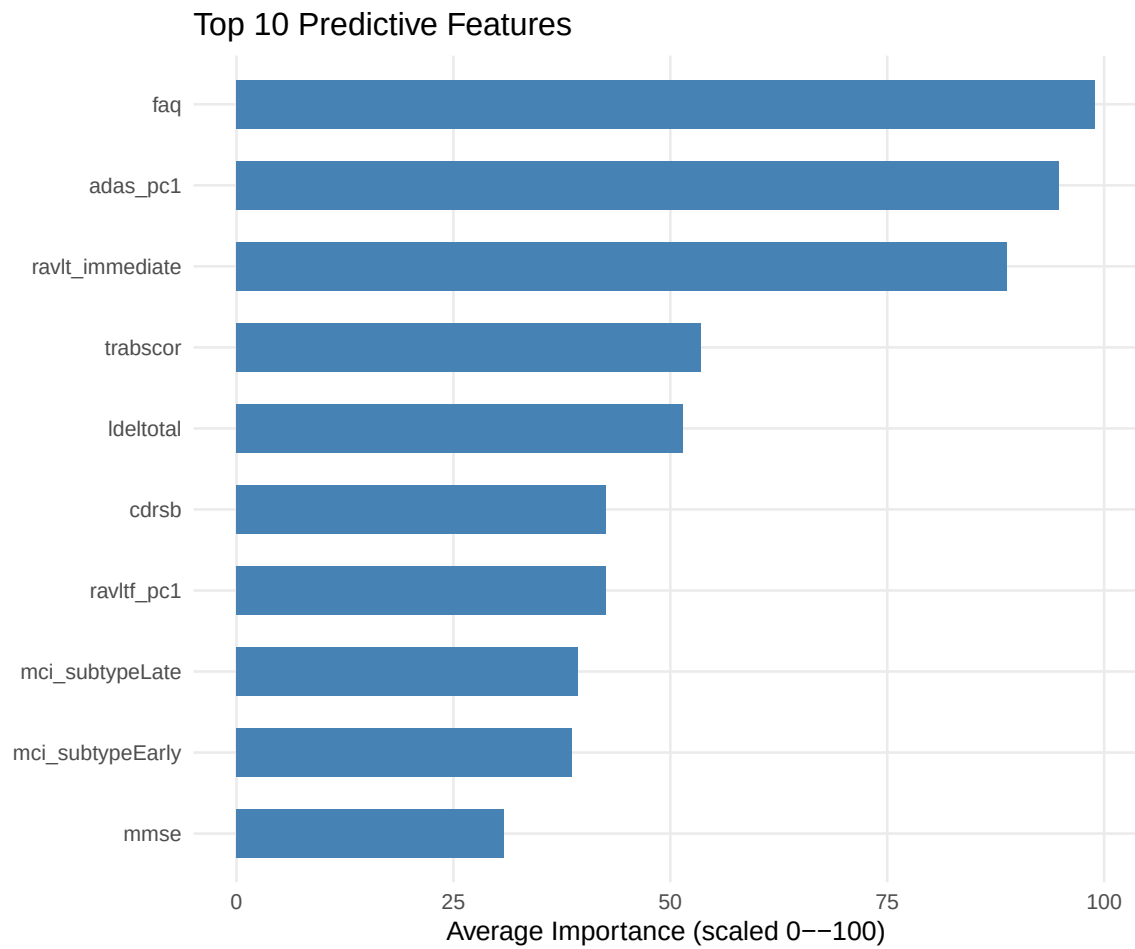


Figure 3: Top 10 predictive features by average importance across models

Table 5: Test set AUROC by MCI subtype

Subtype	N	AUROC		
		LR	RF	XGBoost
Early	74	0.875	0.900	0.871
Late	102	0.855	0.843	0.840

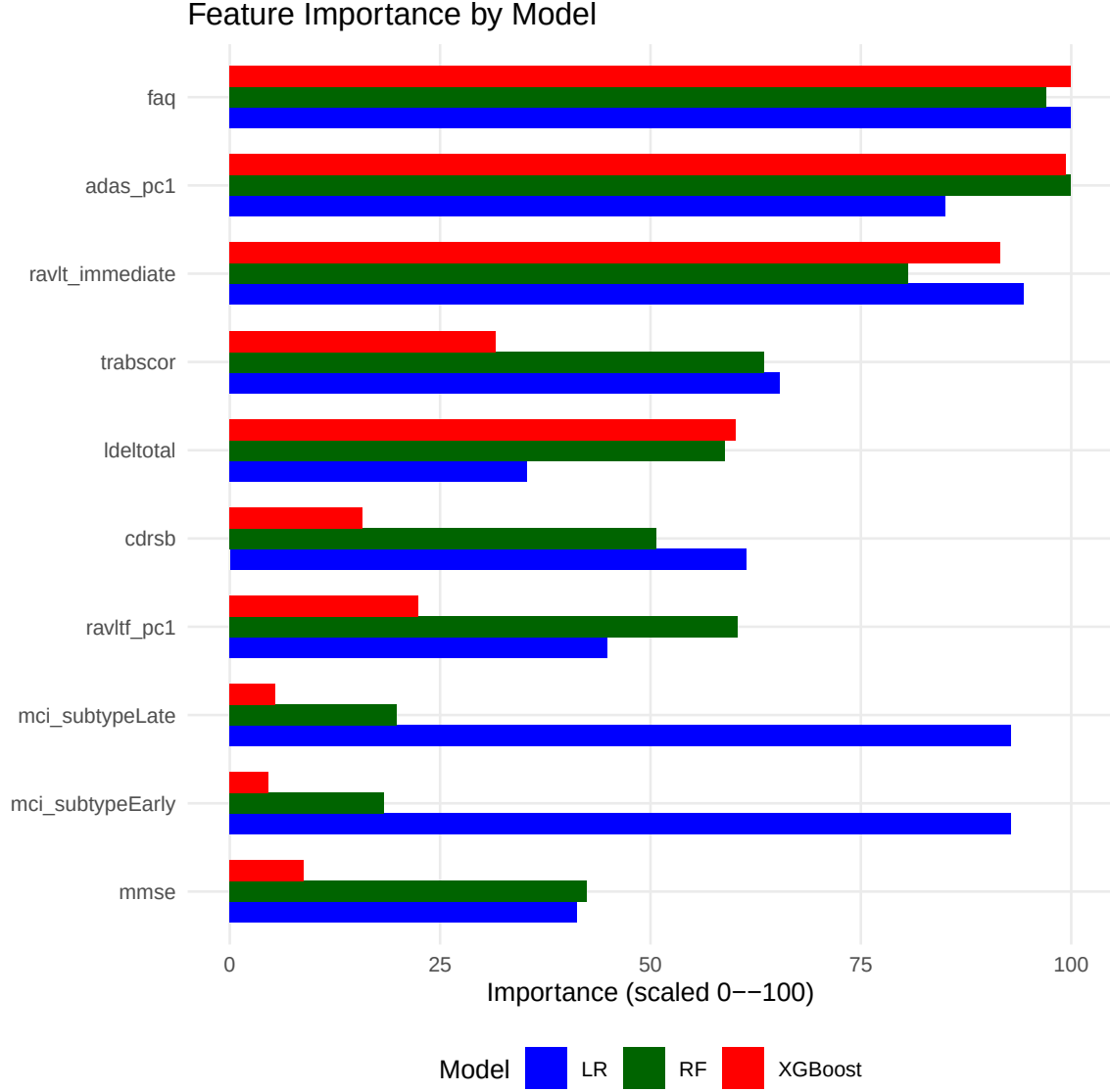


Figure 4: Feature importance rankings across models

Table 6: Sensitivity/specificity at various thresholds for the best model

Table 7: Performance at various sensitivity thresholds (LR model).

Target Sens.	Sensitivity	Specificity	PPV	NPV
100%	100.0%	43.2%	37.0%	100.0%
95%	95.0%	56.8%	NA%	NA%
90%	90.0%	70.5%	NA%	NA%
85%	85.0%	78.8%	NA%	NA%
80%	80.0%	84.8%	NA%	NA%
75%	75.0%	90.9%	73.3%	91.6%

4.7 Sensitivity Analyses

4.7.1 Performance by MCI Subtype

4.7.2 Threshold Analysis

5 Discussion

5.1 Principal Findings

This analysis developed and evaluated three machine learning algorithms for predicting 3-year conversion from MCI to AD using baseline sociodemographic, clinical, and neuropsychological data from the ADNI cohort. All three models achieved discriminative performance, with test-set AUROCs summarized in Table 3. The most consistently important predictors included clinical and cognitive measures that are routinely collected in memory clinic assessments.

5.2 Comparison with Prior Work

Our findings are broadly consistent with those of Grassi et al. (2019), who reported an ensemble AUROC of 0.88 using a similar predictor set. Differences in exact performance may reflect variations in sample composition (ADNI data versions), inclusion criteria, and the use of a held-out test set rather than cross-validation alone. Rye et al. (2022) similarly demonstrated that clinically transferable features provide strong discriminative performance, reinforcing the viability of accessible prediction tools.

5.3 Clinical Implications

The models developed here rely exclusively on clinical and neuropsychological measures that are routinely collected in memory clinic settings. This design choice maximizes clinical translatability: the prediction tool could be deployed without requiring expensive PET imaging or invasive lumbar puncture. The threshold analysis (Table 6) illustrates the sensitivity-specificity trade-off, allowing clinicians to select operating points based on clinical priorities (e.g., high sensitivity for screening, high specificity for trial enrollment).

5.4 Strengths and Limitations

Strengths:

1. Use of well-characterized ADNI cohort with standardized assessments
2. Focus on clinically available predictors for translatability
3. Comparison of multiple machine learning algorithms
4. Rigorous train/test split with cross-validation

Limitations:

1. ADNI participants are predominantly White and well-educated, limiting generalizability to diverse community populations
2. Retrospective design cannot establish causation
3. No external validation cohort
4. Median imputation may attenuate associations for variables with substantial missingness

5.5 Future Directions

External validation in independent, community-based cohorts is needed to establish generalizability. Future work could compare the performance of clinical-only models against models incorporating neuroimaging and fluid biomarkers to quantify the incremental value of expensive assessments. Longitudinal prediction models that incorporate change trajectories, rather than baseline snapshots alone, may further improve prognostic accuracy.

6 Conclusion

Machine learning models using readily available clinical and neuropsychological measures can discriminate MCI subjects at risk for 3-year conversion to AD. The approach prioritizes clinical translatability by excluding imaging and genetic biomarkers, supporting potential deployment in routine memory clinic assessments. External validation and prospective evaluation are necessary before clinical implementation.

7 References

- Donohue, M. C., Hussen, K., Langford, O., Gallardo, R., Jimenez-Maggiora, G., & Aisen, P. S. (2026). Alzheimer’s clinical research data via r packages: The alzverse. *Alzheimer’s & Dementia*, 22, e71152. <https://doi.org/10.1002/alz.71152>
- Grassi, M., Rouleaux, N., Caldirola, D., Loewenstein, D., Schruers, K., Perna, G., & Dumontier, M. (2019). A novel ensemble-based machine learning algorithm to predict the conversion from mild cognitive impairment to alzheimer’s disease using socio-demographic characteristics, clinical information, and neuropsychological measures. *Frontiers in Neurology*, 10, 1–15. <https://doi.org/10.3389/fneur.2019.00756>
- Petersen, R. C. (2004). Mild cognitive impairment as a diagnostic entity. *Journal of Internal Medicine*, 256(3), 183–194. <https://doi.org/10.1111/j.1365-2796.2004.01388.x>

Rye, I., Vik, A., Kocinski, M., Lundervold, A. J., & Lundervold, A. S. (2022). Predicting conversion to alzheimer's disease in individuals with mild cognitive impairment using clinically transferable features. *Scientific Reports*, 12, 15566. <https://doi.org/10.1038/s41598-022-18870-6>

Table 8: L2-regularized logistic regression coefficients at optimal lambda

Feature	Coefficient
faq	0.3797
ravlt_immediate	-0.3609
mci_subtypeLate	0.3562
mci_subtypeEarly	-0.3562
adas_pc1	-0.3302
ptmarryNever.married	0.3142
ptmarryDivorced	-0.2957
trabscor	0.2650
cdrsb	0.2517
ravltf_pc1	0.1972
mmse	-0.1850
ldeltotal	-0.1656
pteducat	0.1187
age	-0.1068
ptmarryWidowed	0.0655
sexFemale	0.0534
sexMale	-0.0533
ptmarryMarried	0.0515
ravlt_learning	-0.0484

8 Supplementary Material

8.1 Supplementary Table S1: Logistic Regression Coefficients

8.2 Supplementary Figure S1: Calibration Plots

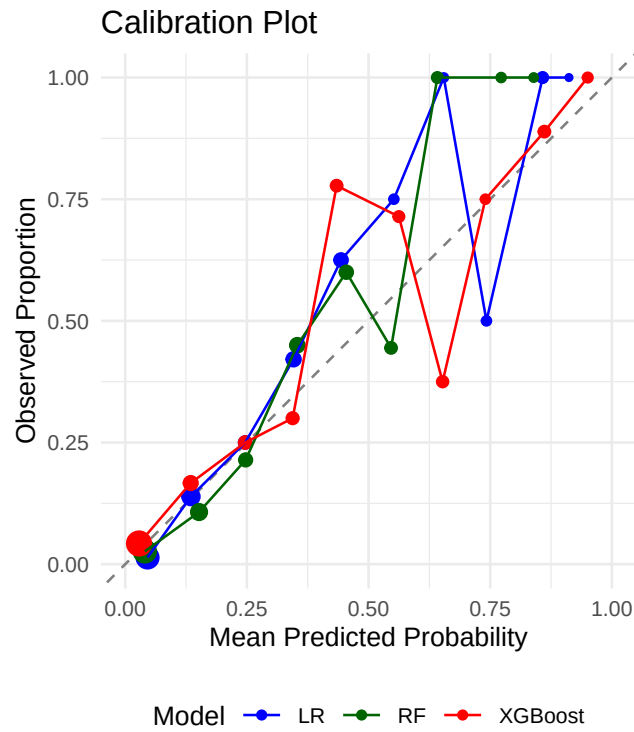


Figure 5: Calibration plots: predicted vs. observed conversion probability

8.3 Supplementary Figure S2: Cross-Validation Performance

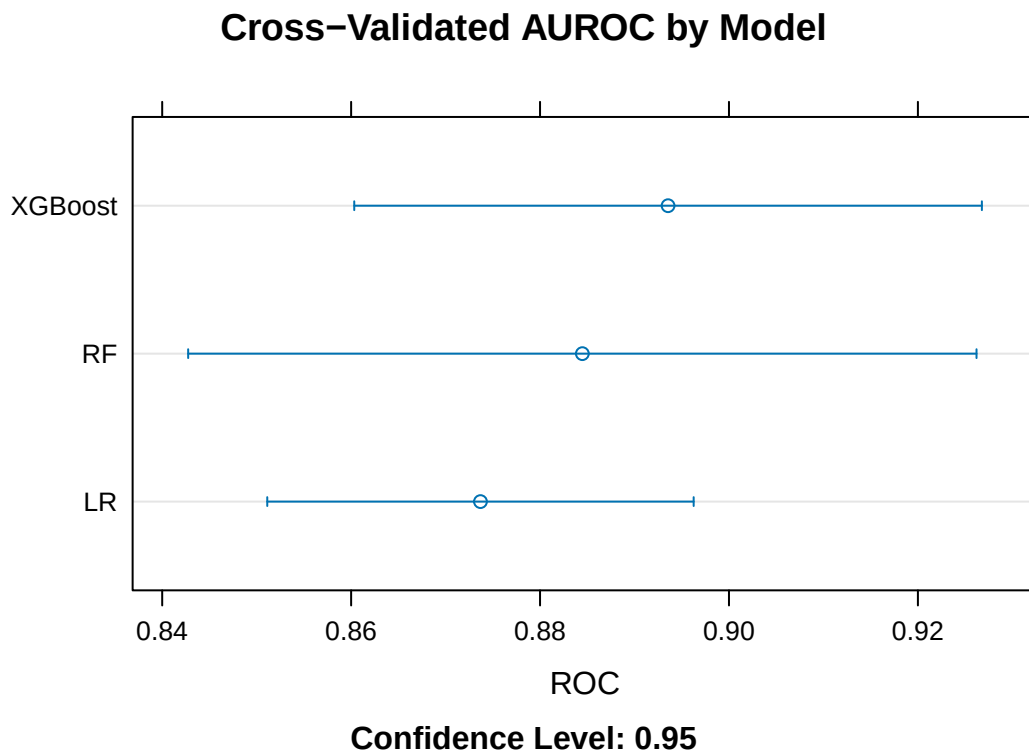


Figure 6: Cross-validated AUROC distributions across 5 folds

9 Session Information

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sessionInfo()
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LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; LAPACK
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attached base packages:
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other attached packages:

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[7] lattice_0.22-7     kableExtra_1.4.0  knitr_1.50
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